

Math_31B_22W_Discussion_Logistics

Tuesday, January 4, 2022 9:28 AM



Math_31B...

1130-1230

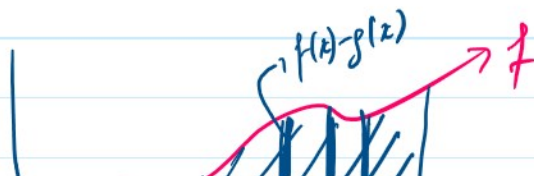
Math 31B Discussion Section (1A and B) Information/Logistics

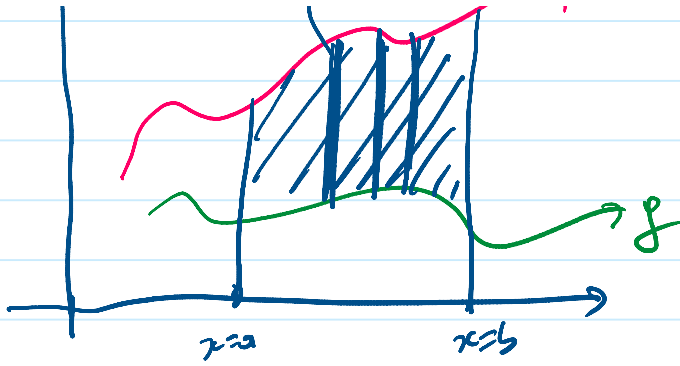
1. Instructor: Dr. Paige Greene
2. Teaching Assistant: Osman Akar
3. Discussion Times and Links (all times in PST).
 - (a) **1A:** Tuesdays 10-10:50am at Geology Building 4645
<https://ucla.zoom.us/j/92643693231?pwd=bzRvV2pRk3Q5anlVbXg2YXVBQTU09>
 Meeting ID: 926 4369 3231, Passcode: 967460
 - (b) **1B:** Thursdays 10-10:50am at Franz Hall 2258A
<https://ucla.zoom.us/j/91322972597?pwd=ZmJHOEg0O1VlSUhDT3VON2EwWTNlUT09>
 Meeting ID: 913 2297 2597, Passcode: 093902
4. **Office Hours:** You are most welcomed to attend the office hours, even when you don't have questions but you want to see other student's questions, or you just want to chat in general. I was also UCLA undergrad and I have spent almost 7 years here, so I have quite an experience of being a math major.
 I plan to hold both in person and online office hours (via zoom of course), but this is subject to change. Of course the first two weeks will be via zoom.
 - **Zoom OH:** Time: TBD. Zoom link is the same as the open OH link. 11am-12pm?
 - **Open OH:** 9-10am on Tuesdays. Here is the zoom link details:
<https://ucla.zoom.us/j/97532843825?pwd=RldkOEISd25Le0pYT3RlNTgrbDBKQT09>
 Meeting ID: 975 3284 3825, Passcode: 896714
5. Email: oak@math.ucla.edu.
6. Email Policy: Please do not send me math questions via email, instead I prefer you to come to my office hours. It is quite time consuming to answer math questions via email because I cannot write mathematical symbols easily.
7. As you all know, the first two weeks will be online via zoom. When we get back in person classes, I still plan to broadcast and record the live discussion sessions, and upload the recording to the canvas – as long as the technology allows me to do it. In other words, undeniably there will be technical errors throughout the quarter, so safest way to attend discussion is in person, when we come back to the campus.

8. Useful Links (Free materials UCLA offers):

- (a) MATLAB is free for UCLA Students: <https://softwarecentral.ucla.edu/matlab-getmatlab>
- (b) Microsoft Office is also free: <https://www.it.ucla.edu/news/microsoft-office-proplus>
- (c) You can read *The Economist* magazine for free if you are on UCLA network. If not, you can use UCLA VPN. See here for more info <https://www.anderson.ucla.edu/rosenfeld-library/databases/business-databases-by-name/economist>. Note that being on UCLA network is also useful to download articles from a number of publishers and also from UCLA library (NOTE: for some reason this free access does not seem to be working for a couple of weeks, but I hope they will fix it soon).

ch 6-1 Area between two Curves.





f, g are functions

$$\text{Area of the region} = \int_a^b f(x) - g(x) dx$$

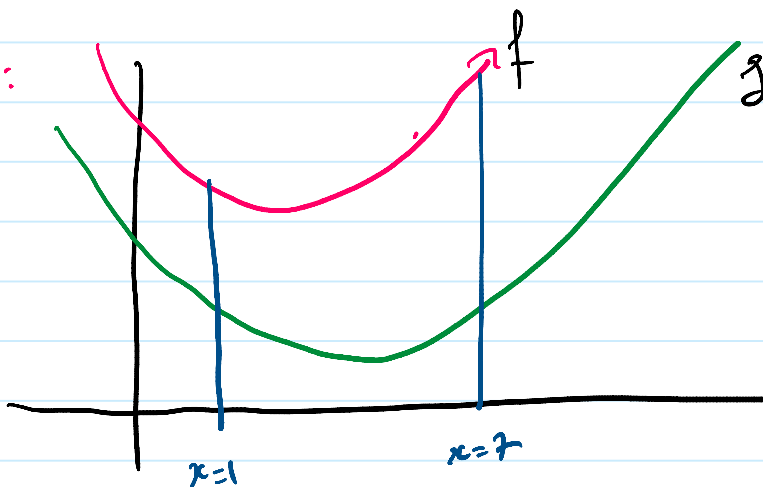
Ex 1: Define $f(x) = 2x^2 - 3x + 4$ and $g(x) = x^2 - 4x + 5$

Find the area of the region between f & g from

$x=1$ to $x=7$.

A) 10 B) 18 C) 32 D) 132 E) None. F)

Sol:



$$f(x) - g(x) = (2x^2 - 3x + 4) - (x^2 - 4x + 5)$$

$$= x^2 + x - 1 > 0$$

Note that for $1 \leq x \leq 7$, $x^2 + x - 1 > 0$

Indeed, f lies above g between $x=1$ and $x=7$.

$$A = \int_1^7 x^2 + x - 1 \, dx = \left. \frac{x^3}{3} + \frac{x^2}{2} - x \right|_{x=1}^7$$

Recall, $\int x^n = x^{n+1} \cdot \frac{1}{n+1} + C$

$$= \left(\frac{7^3}{3} + \frac{7^2}{2} - 7 \right) - \left(\frac{1^3}{3} + \frac{1^2}{2} - 1 \right)$$

$$= \frac{7^3 - 1^3}{3} + \frac{7^2 - 1^2}{2} - 7 + 1$$

$$= \frac{342}{3} + \frac{48}{2} - 6 = 114 + 24 - 6 = 132$$

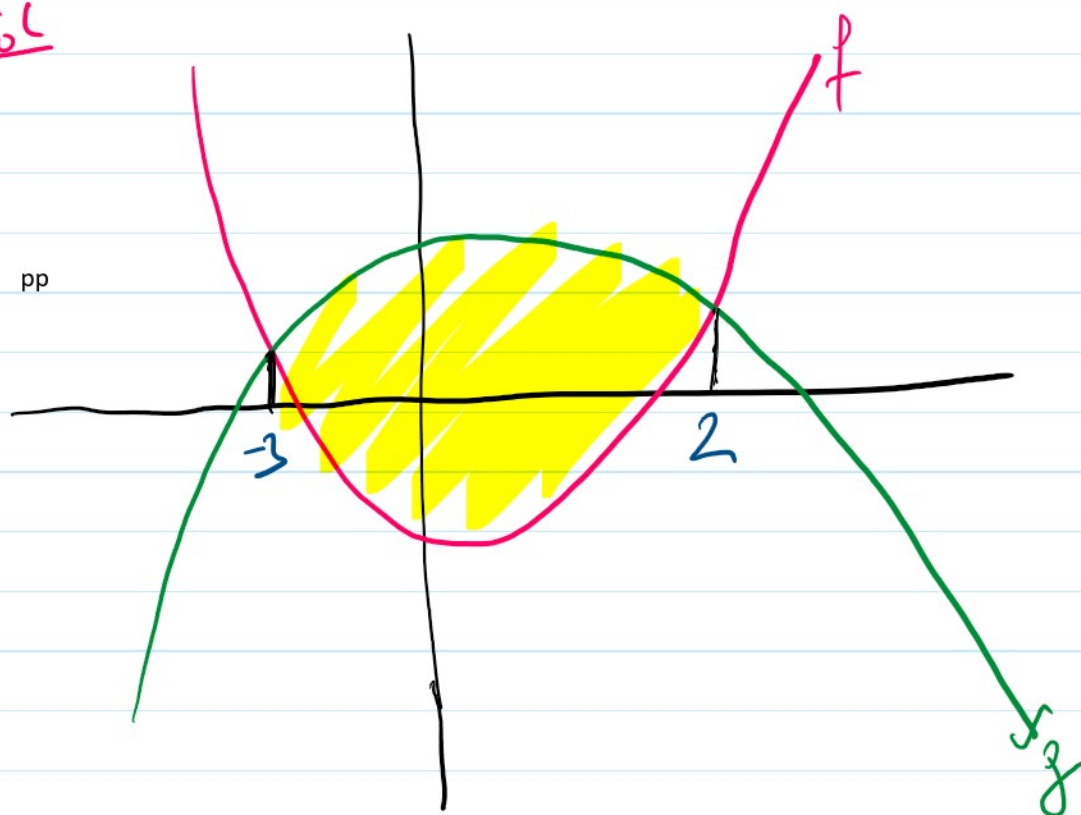
Ex 2 $f(x) = \underline{2}x^2 + 3x - 11$

$$g(x) = \underline{-4}x^2 - 3x + 25$$

when you draw both of the parabolas, it's gonna create a closed region. Find the area of this region.

A) 125 B) 132 C) 171 D) 240 E) None.

Sol



a/b are the points when $f(x) = g(x)$

$$f(x) - g(x) = 0$$

$$2x^2 + 3x - 11 - (-4x^2 - 3x + 25) =$$

$$= 6x^2 + 6x - 36 = 6(x^2 + x - 6)$$

$$= 6(x-2)(x+3)$$

$$f(x) - f(x) = 6(x-2)(x+3) = 0 \quad \text{at } x = -3, x = 2.$$

$$A = \int_{-3}^2 g(x) - f(x) \, dx$$

$$= \int_{-3}^2 -6(x^2 + x - 6) \, dx$$

$$= \int_{-3}^2 -6x^2 - 6x + 36 \, dx = -2x^3 - 3x^2 + 36x \Big|_{x=-3}^2 =$$

$$= (-2 \cdot 2^3 - 3 \cdot 2^2 + 36 \cdot 2) - (-2(-3)^3 - 3(-3)^2 + 36(-3))$$

$$= (-16 - 12 + 72) - (54 - 27 - 108)$$

$$= \boxed{1125}$$